STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-GOOD-QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3036-3050 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3052-3095, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 15 of queue_sdga.json (pb=1894, stmt 3036-3050, proof 3052-3095).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-good-quotient' also exists in stacks-dga — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FSC: sources/stacks-sdga/pinned-original-source.tex, source lines 2983–2991. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSD: sources/stacks-sdga/pinned-original-source.tex, source lines 3022–3029. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FSB: sources/stacks-sdga/pinned-original-source.tex, source lines 2891–2897. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FS8: sources/stacks-sdga/pinned-original-source.tex, source lines 2785–2804. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. Let $\mathcal{M}$ be a differential graded $\mathcal{A}$-module. There exists a homomorphism $\mathcal{P} \to \mathcal{M}$ of differential graded $\mathcal{A}$-modules with the following properties*

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $\mathcal{P} \to \mathcal{M}$ *is surjective,*
(2) $\mathrm{Ker}(d_\mathcal{P}) \to \mathrm{Ker}(d_\mathcal{M})$ *is surjective, and*
(3) $\mathcal{P}$ *is good.*

Proof. Consider triples (U, k, x) where U is an object of $\mathcal{C}$, $k \in \mathbf{Z}$, and x is a section of $\mathcal{M}^k$ over U with $\mathrm{d}_\mathcal{M}(x) = 0$. Then we obtain a unique morphism of differential graded $\mathcal{A}_U$-modules $\varphi_x : \mathcal{A}_U[-k] \to \mathcal{M}|_U$ mapping 1 to x. This is adjoint to a morphism $\psi_x : j_{U!}\mathcal{A}_U[-k] \to \mathcal{M}$. Observe that $1 \in \mathcal{A}_U(U)$ corresponds to a section $1 \in j_{U!}\mathcal{A}_U[-k](U)$ of degree k whose differential is zero and which is mapped to x by ψ_x. Thus if we consider the map

$$\bigoplus\nolimits_{(U,k,x)} j_{U!}\mathcal{A}_U[-k] \longrightarrow \mathcal{M}$$

then we will have conditions (2) and (3). Namely, the objects $j_{U!}\mathcal{A}_U[-k]$ are good (Lemma 0FSB) and any direct sum of good objects is good (Lemma 0FSD).

Next, consider triples (U, k, x) where U is an object of $\mathcal{C}$, $k \in \mathbf{Z}$, and x is a section of $\mathcal{M}^k$ (not necessarily annihilated by the differential). Then we can consider the cone C_U on the identity map $\mathcal{A}_U \to \mathcal{A}_U$ as in Remark 0FS8. The element x will determine a map $\varphi_x : C_U[-k-1] \to \mathcal{A}_U$, see Remark 0FS8. Now, since we have an admissible short exact sequence

$$0 \to \mathcal{A}_U \to C_U \to \mathcal{A}_U[1] \to 0$$

we conclude that $j_{U!}C_U$ is a good module by Lemma 0FSC and the already used Lemma 0FSB. As above we conclude that the direct sum of the maps $\psi_x : j_{U!}C_U \to \mathcal{M}$ adjoint to the φ_x

$$\bigoplus\nolimits_{(U,k,x)} j_{U!}C_U \longrightarrow \mathcal{M}$$

is surjective. Taking the direct sum with the map produced in the first paragraph we conclude. □